Composition

Given the relations R and S, let us define a new relation. Suppose that $(a,b) \in S$ and $(b,c) \in R$. Therefore, a is related to c through b. The new relation will contain the ordered pair (a,c) to represent this relationship.

Definition:

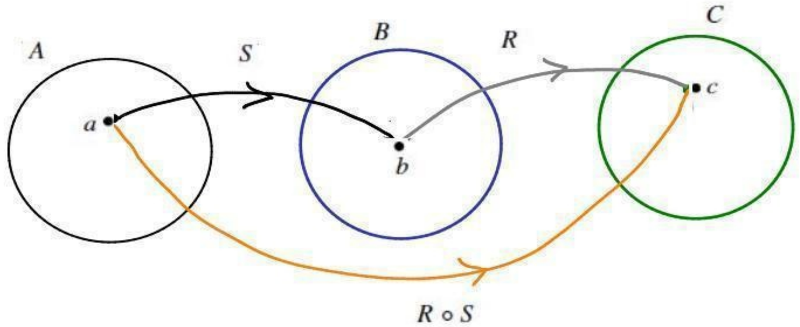
Let $S \subseteq A \times B$ and $R \subseteq B \times C$. The **composition** of R and S is the subset of $A \times C$ defined as

$$R \circ S = \{(x,z) : \exists y \text{ such that } y \in B \wedge (x,y) \in S \wedge (y,z) \in R\}.$$

As illustrated in Figure below the reason that $(a,c) \in R \circ S$ is because $(a,b) \in S$ and $(b,c) \in R$. That is, a is related to c via $R \circ S$ because a is related to b via S and then b is related to c via R. The composition can be viewed as the direct path from a to c that does not require the intermediary b.

$S \subseteq A \times B$

$R \subseteq B \times C$



$R \circ S \subseteq A \times C$

A composition of relations.

$a \; R \circ S \; c \iff \exists b \in B,\ a S b \wedge b R c$

i.e $(a,c) \in R \circ S \iff \exists b \in B,\ (a,b) \in S \wedge (b,c) \in R.$

Example 1:

$R \circ S = \{(0,3), (0,4), (0,5)\}$

To clarify the definition, let $S \circ R = \{\ \} = \emptyset$

$$R = \{(2,4), (1,3), (2,5)\}$$

$2R4,\ 1R3,\ 2R5$

and

$0S1,\ 1S0,\ 0S2,\ 2S0$

$$S = \{(0,1), (1,0), (0,2), (2,0)\}.$$

We have that $R \circ S = \{(0,3), (0,4), (0,5)\}$. Notice that $(0,3) \in R \circ S$ because $(0,1) \in S$ and $(1,3) \in R$. However, $S \circ R$ is empty since ran(R) and dom(S) are disjoint.

Therefore, $S \circ R \neq R \circ S$.

Example 2:

Define

$$\mathcal{R} = \{(x, y) \in \mathbf{R}^2 : x^2 + y^2 = 1\}$$

and

$$S = \{(x, y) \in \mathbf{R}^2 : y = x + 1\}.$$

Notice that R is the unit circle and S is the line with slope of 1 and y-intercept of $(0, 1)$. Let us find $R \circ S$.

$$\begin{aligned}\mathcal{R} \circ S &= \{(x, z) \in \mathbf{R}^2 : \exists y \text{ such that } y \in \mathbf{R} \wedge (x, y) \in S \wedge (y, z) \in \mathcal{R})\} \\ &= \{(x, z) \in \mathbf{R}^2 : \exists y \text{ such that } y \in \mathbf{R} \wedge y = x + 1 \wedge y^2 + z^2 = 1)\} \\ &= \{(x, z) \in \mathbf{R}^2 : (x + 1)^2 + z^2 = 1\}.\end{aligned}$$

Therefore, $\mathcal{R} \circ S$ is the circle with center $(-1, 0)$ and radius 1.

Theorem 1:

If $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$, then $T \circ (S \circ R) = (T \circ S) \circ R$.

Proof:
Assume that $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Then,

$$\begin{aligned}&(a, d) \in T \circ (S \circ R) \\ &\Leftrightarrow \exists c, c \in C \wedge (a, c) \in S \circ R \wedge (c, d) \in T) \\ &\Leftrightarrow \exists c, c \in C \wedge \exists b, b \in B \wedge (a, b) \in R \wedge (b, c) \in S) \wedge (c, d) \in T) \\ &\Leftrightarrow \exists c \exists b, c \in C \wedge b \in B \wedge (a, b) \in R \wedge (b, c) \in S \wedge (c, d) \in T) \\ &\Leftrightarrow \exists b \exists c, b \in B \wedge (a, b) \in R \wedge c \in C \wedge (b, c) \in S \wedge (c, d) \in T) \\ &\Leftrightarrow \exists b, b \in B \wedge (a, b) \in R \wedge \exists c, c \in C \wedge (b, c) \in S \wedge (c, d) \in T) \\ &\Leftrightarrow \exists b, b \in B \wedge (a, b) \in R \wedge (b, d) \in T \circ S) \\ &\Leftrightarrow (a, d) \in (T \circ S) \circ R. \blacksquare\end{aligned}$$

Definition. Let $R \subseteq A \times B$ be a relation. Then, its *inverse* $R^{-1} \subseteq B \times A$ is the set

$$R^{-1} = \{(y, x) \in B \times A : (x, y) \in R\}.$$

To find the elements of R^{-1}, you simply switch the components of each ordered pair in R.

Example 01. Let $R = \{(1,3), (2,2), (2,3), (3,2), (4,1), (5,2)\}$.
Then, $R^{-1} = \{(3,1), (2,2), (3,2), (2,3), (1,4), (2,5)\}$.

Example 02. The less-than relation on $\mathbf{Z}$ is defined as

$$L = \{(a, b) : a, b \in \mathbf{Z} \wedge a < b\}.$$

Then,

$$\begin{aligned}L^{-1} &= \{ (y, x) \in \mathbf{Z}^2 : (x, y) \in L\} \\ &= \{(y, x) \in \mathbf{Z}^2 : x < y\} \\ &= \{(y, x) \in \mathbf{Z}^2 : y > x\}.\end{aligned}$$

This shows that less-than and greater-than are inverse relations.

Ex.

01. Let $R \subseteq \{1,2,3,4\} \times \{1,2,3,4\}$ be the relation $R = \{(1,3),(1,4),(2,2),(2,4),(3,1),(3,2),(4,4)\}$.
Compute R^{-1}.

02. For the relation $R = \{(x,y) : x \leq y\}$ defined on $\mathbb{N}$, what is R^{-1}?

03. *Given any relations $R, S \subseteq A \times A$:*

1. $(R^{-1})^{-1} = R$
2. $R \subseteq S \iff R^{-1} \subseteq S^{-1}$
3. $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$
4. $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$

Ex

Consider $R = \{(2,1),(4,3)\}$, which is a relation on $\{1,2,3,4\}$.
Then, $R^{-1} = \{(1,2),(3,4)\}$, and we see that

$$R \circ R^{-1} = \{(1,1),(3,3)\}$$

and

$$R^{-1} \circ R = \{(2,2),(4,4)\}$$

Consider $S = \{(2,1),(2,3),(4,3)\}$.

Then, we have $S^{-1} = \{(1,2),(3,2),(3,4)\}$

but

$$S \circ S^{-1} = \{(1,1),(1,3),(3,1),(3,3)\}$$

and

$$S^{-1} \circ S = \{(2,2),(2,4),(4,4)\}.$$

Note that, $\{(1,1),(3,3)\} \subseteq S \circ S^{-1}$ and $\{(2,2),(4,4)\} \subseteq S^{-1} \circ S$.

This can be generalized.

Definition. For any set A, $I_A = \{(a,a) : a \in A\}$ is called the **identity** on A.

Example: The identity on $\mathbf{R}$ is $I_{\mathbf{R}} = \{(x,x) : x \in \mathbf{R}\}$, and the identity on $\mathbf{Z}$ is $I_{\mathbf{Z}} = \{(x,x) : x \in \mathbf{Z}\}$. Notice that $\varnothing$ is the identity on $\varnothing$.

Ex.

Let $R \subseteq A \times B$.

(a) Prove $R \circ I_A = R$ and $I_B \circ R = R$.

(b) Show that if there exists a set C such that A and B are subsets of C, then $R \circ I_C = I_C \circ R = R$. Ex

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